Extended Abstract

Motivation Imitation learning, especially behavior cloning, has become a dominant approach for learning robotic skills (Mandlekar et al., 2021; Zhao et al., 2023; Chi et al., 2023; Black et al., 2024b; Jiang et al., 2025b). However, training these policies requires large-scale human demonstrations (Ebert et al., 2021; Brohan et al., 2022b; Walke et al., 2023; Khazatsky et al., 2024), which are costly to collect. In contrast, physics-based simulation can generate unlimited data (Liu and Negrut, 2021), potentially alleviating this bottleneck. Motivated by the question: *Can synthesized robot data help learn effective policies?*, we develop a simulation-based data synthesis pipeline for contact-rich manipulation and evaluate its utility for learning competent policies.

Method We propose a data synthesis pipeline using constrained motion planning for the contact-rich task of table wiping (Li et al., 2024), where the robot must maintain contact with the surface while avoiding obstacles. The pipeline comprises four steps: **1)** First, we solve for *collision-free* inverse kinematics (IK) poses on the table surface. Collision avoidance is important for household tasks due to cluttered distributions of daily objects. We discard poses near obstacles like walls. **2)** Second, two collision-free IK poses are sampled as start and goal configurations. We impose two constraints for planning — one confines the robot hand orientation orthogonal to the table surface, and the other limits the hand height to ensure surface contact. **3)** Third, using cuRobo (Sundaralingam et al., 2023), we plan minimal-jerk, collision-free trajectories satisfying the constraints. If planning fails, the process restarts. **4)** Finally, the planned trajectory is executed in simulation. Successful executions are saved as demonstrations. To achieve a high throughput for data synthesis, we parallelize the simulator instances. After generating a target amount of demonstrations, we use them to train imitation learning policies, including ACT (Zhao et al., 2023) and Diffusion Policy (DP, Chi et al. (2023)).

Implementation Our synthesis pipeline is implemented using cuRobo within OmniGibson (Li et al., 2024), built on NVIDIA IsaacSim [1]. We use the `Rs_int` indoor scene[2] and a mobile bimanual robot (Jiang et al., 2025b). We generate 100,000 demonstrations. Policies are trained as state-based, goal-conditioned models using robot proprioception and 7-DoF wiping goals. Actions are 22-DoF joint positions, 3 for the mobile base, 7 for each arm and gripper, and 4 for the torso. We compare three implementations of imitation learning policies — ACT (Zhao et al., 2023), UNet-based DP (Chi et al., 2023), and DP with a transformer backbone similar to Jiang et al. (2025b).

Results We report two quantitative metrics: *success rate* and *contact ratio*, where success rate is computed over 200 evaluation rollouts (higher is better) and contact ratio counts the percentage of timesteps when the robot hand is in contact with the table surface (higher is better). Quantitative results are summarized in Table 1. Qualitative results are visualized in Fig. 4.

Table 1: **Summary of quantitative results.**

Method	ACT	DP	DP w/ Transformer
Success Rate	68%	56%	66%
Contact Ratio	71%	64%	63%

Discussion All three policies trained on synthetic data complete the task with reasonable success and maintain hand contact for over half of the trajectory. This supports the utility of simulation-based synthetic data for learning contact-rich behaviors.

Conclusion, Limitations, and Future Work We introduce a method for synthesizing robot demonstrations via constrained motion planning for a table-wiping task. Policies trained on 100,000 such demonstrations perform the task effectively, highlighting the viability of synthetic data for imitation learning. However, our method still has limitations in sim-to-real transfer and the manual specification of task-specific constraints. Future work could focus on automating the constraints construction using LLMs, and tackling the sim-to-real transfer problem.

[1] `https://developer.nvidia.com/isaac/sim`

[2] `https://behavior.stanford.edu/omnigibson/modules/scenes.html`

Robotic Data Synthesis using Constrained Motion Generation for Robot Learning

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Abstract

We present a data synthesis pipeline for training robot manipulation policies using constrained motion generation. Focusing on a contact-rich table-wiping task, our method generates trajectories that maintain surface contact and avoid collisions by solving for feasible inverse kinematics poses and planning constrained, minimal-jerk motions using cuRobo (Sundaralingam et al., 2023). We synthesize 100,000 demonstrations in the OmniGibson (Li et al., 2024) simulator and train goal-conditioned imitation learning policies, including ACT (Zhao et al., 2023) and Diffusion Policy variants (Chi et al., 2023; Jiang et al., 2025b). Evaluated in simulation, these policies achieve strong success rates and high table-robot contact ratios, demonstrating the effectiveness of synthetic data for learning contact-rich robot behaviors.

1 Introduction

Learning from human demonstrations is a powerful paradigm for teaching robots complex manipulation skills. A common approach for collecting such data is teleoperation, where a human directly controls the robot to demonstrate desired behaviors. When scaled up, teleoperated data has enabled training visuomotor policies with impressive generalization and success in challenging manipulation tasks Brohan et al. (2022a); O'Neill et al. (2024); Khazatsky et al. (2024); Black et al. (2024a); Intelligence et al. (2025). However, this data collection process remains expensive and time-consuming, especially for tasks that require high-quality demonstrations to ensure effective policy learning.

Recently, collecting a small amount of human teleoperation data and then synthesizing additional data in simulation has become a popular approach to scale up data collection Mandlekar et al. (2023); Garrett et al. (2024); Jiang et al. (2025a); Xue et al. (2025); Yang et al. (2025). Compared to offline data augmentation techniques such as those based on image augmentation Laskin et al. (2020); Yarats et al. (2021); Pitis et al. (2020); Yu et al. (2023); Chen et al. (2023), this approach can autonomously generate new behaviorally diverse data for the same task. This process enlarges the support and convergence region of the policies and reduces the teacher-student distribution mismatch Ross et al. (2011) through new generated experiences validated in simulation to ensure quality. Notably, the X-Gen family of techniques Mandlekar et al. (2023); Garrett et al. (2024); Jiang et al. (2025a); Xue et al. (2025); Yang et al. (2025) leverages simulation and augmentation based on a small number of human demonstrations that are used as seeds to generate multiple new variations automatically. While they have shown success, they still require human to collect the initial seeding data and are limited to simple table-top pick-and-place tasks.

In this project, we develop a robotic data synthesis method in simulation using constrained motion generation without the need of initial human demonstrations and apply it to a complex mobile manipulator robot. We focus on a contact-rich table-wiping task and use our data synthesis method to generate 100,000 successful trajectories. Imitation learning policies trained on this synthetic data

Figure 1: **Visualizations of the robot and scene and setup.** From left to right: 1) The chosen indoor apartment-like scene; 2) Top-down view of the scene; 3) Our robot and task setup.

can autonomously complete the task with success rates up to 68%, verifying the effectiveness and helpfulness of our data synthesis method.

2 Related Work

Data Acquisition for Robot Learning. Collecting large-scale human teleoperation data for robot learning incurs considerable costs. Scaling up data collection requires a large number of human operators over extended periods of time Brohan et al. (2022a); O'Neill et al. (2024); Khazatsky et al. (2024) or needs to rely on crowdsourced teleoperation systems such as RoboTurk Mandlekar et al. (2018). Offline data augmentation techniques can boost data quantity and diversity by perturbing existing trajectories Mandlekar et al. (2023); Garrett et al. (2024), or leveraging image augmentation techniques and generative models Laskin et al. (2020); Yarats et al. (2021); Pitis et al. (2020); Yu et al. (2023); Chen et al. (2023). However, the augmented data may not always be executable by real robots. A promising alternative is to leverage automated data generation and validation in simulation. Fully automated approaches include trial-and-error Levine et al. (2018); Pinto and Gupta (2016); Yu et al. (2016); Dasari et al. (2019) and pre-programmed (e.g., scripted) experts James et al. (2020); Gu et al. (2023); Jiang et al. (2022); Wang et al. (2023), which are yet to be proven effective for complex tasks without additional components such as planning Dalal et al. (2023). X-Gen Mandlekar et al. (2023); Garrett et al. (2024); Jiang et al. (2025a); Xue et al. (2025); Yang et al. (2025) represents a hybrid approach that leverages simulation and augmentation to build upon a handful of human demonstrations as seeds to generate many new variations, augmenting the data by a factor of $25\times$ to $350\times$ Mandlekar et al. (2023); Jiang et al. (2025a), while ensuring synthesized data is valid. Despite their effectiveness, they still rely on human to collect initial demonstrations as the *seeding* data. Instead, our proposed method is fully automatic and bypasses the requirements on seeding demonstrations.

Robot Policy Learning from Synthetic Data To leverage synthetic data for robot policy training, previous work primarily focuses on sim-to-real transfer. For example, Jiang et al. (2024) first trains robot manipulation policies in simulation using reinforcement learning, then deploys the simulation policies in the real world by learning from human correction. Dai et al. (2024) trains robot manipulation policies in simulation by training on object assets that are close to but not identical to real-world objects—the so called digital cousins. However, both of them only demonstrate on single-arm robots. In our project, we synthesize data to learn a humanoid robot to perform contact-rich wiping table task.

3 Robotic Data Synthesis using Constrained Motion Generation

In this section, we introduce our proposed robotic data synthesis method by using constrained motion generation. We first overview the robot and scene setup in the simulation, then give details about our data synthesis method. Next, we present policy learning methods from our synthetic data. Finally, we present critical implementation details.

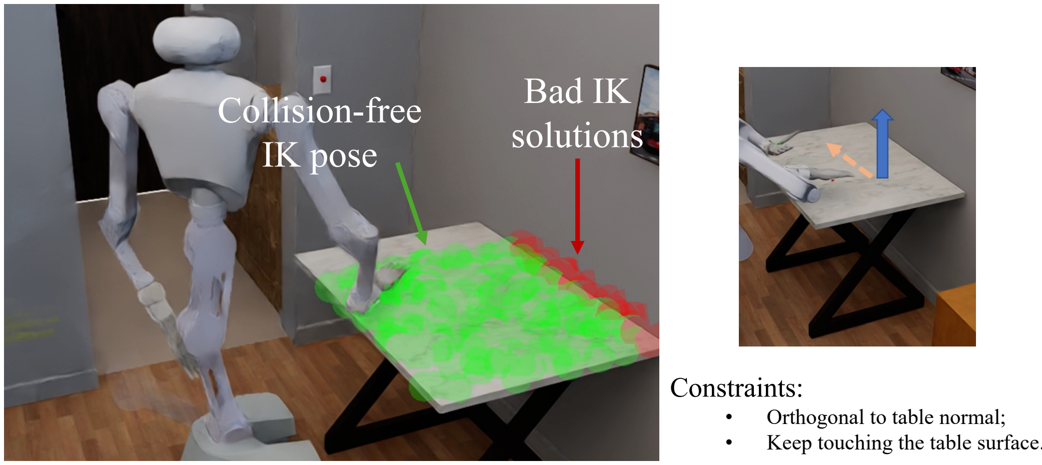

Figure 2: **Data synthesis initial poses and constraints.** Left: Collision-free IK poses (represented as green dots) and collision IK poses (represented as red dots). Right: Two constraints used in motion planning.

3.1 Robot and Scene Setup

We choose the contact-rich task of table wiping from Li et al. (2024) because of its commonness and importance among household tasks. We use the `Rs_int` scene[3], which is an indoor apartment-like environment as visualized in Fig. 1. We use a bimanual mobile robot with a 4-DoF torso (Jiang et al., 2025b) and reset it close to the table. As shown in Fig. 1, the robot needs to move its hand to the target wiping position (marked as yellow) while keeps contact with the table surface.

3.2 Data Synthesis Pipeline

Our proposed data synthesis pipeline comprises four steps: **1)** First, we solve for *collision-free* inverse kinematics (IK) poses on the table surface (Fig. 2 where green dots for collision-free IK poses). Collision avoidance is important for household tasks due to cluttered distributions of daily objects. We discard poses near obstacles like walls. **2)** Second, two collision-free IK poses are sampled as start and goal configurations. We impose two constraints for planning — one confines the robot hand orientation orthogonal to the table surface, and the other limits the hand height to ensure surface contact. These two constraints ensure the robot moves its hand in a way that can effectively wipe the table, as demonstrated in Fig. 2. **3)** Third, using cuRobo (Sundaralingam et al., 2023), we plan minimal-jerk, collision-free trajectories satisfying the constraints. If planning fails, the process restarts. **4)** Finally, the planned trajectory is executed in simulation. Successful executions are saved as demonstrations. An example of the generated robot trajectory is visualized in Fig. 3. To achieve a high throughput for data synthesis, we parallelize the simulator instances. We use this pipeline to generate 100,000 successful trajectories in 3 days.

3.3 Policy Learning

We use our synthetic data to train imitation learning policies. We consider three models, the Action Chunking Transformer (Zhao et al., 2023) (ACT), the UNet-based Diffusion Policy (Chi et al., 2023) (DP), and a DP with the transformer backbone similar to Jiang et al. (2025b). We implement all three policies as state-based, goal-conditioned models. They take inputs of the 22-dimensional robot proprioception and a 7-dimensional vector representing the wiping target poses. The output actions are 22-dimensional: 3 for the mobile base, 7 for each arm and gripper, and 4 for the torso. For ACT, we optimize the L2 loss between predicted and ground-truth actions. For DP, we optimize the noise prediction loss where actions are modeled through the diffusion process.

[3] `https://behavior.stanford.edu/omnigibson/modules/scenes.html`

Figure 3: **An example of the generated trajectory.**

3.4 Implementation Details

Our data synthesis method is implemented in the OmniGibson (Li et al., 2023) simulator based on NVIDIA Isaac Sim. We implement all policy models with PyTorch and train them using the PyTorch-Lightning framework. All data synthesis, model training, and policy evaluation are performed on a workstation with NVIDIA RTX4090 GPUs.

4 Experimental Setup

We train all three policies on the generated 100,000 trajectories. During training, we periodically run evaluation consisting of 200 rollouts. We train until the evaluation success rate does not further improve, and select the policy checkpoint corresponding to the highest success rate. A rollout is defined as successful if the robot hand achieves within 5 cm with the target wiping pose in 300 simulation steps. We run the simulator at 30 Hz. We calculate and report two quantitative metrics: *success rate* and *contact ratio*, where success rate is computed over 200 evaluation rollouts (higher is better) and contact ratio counts the percentage of timesteps when the robot hand is in contact with the table surface (higher is better).

5 Results

5.1 Quantitative Evaluation

We summarize the quantitative evaluation results in Table 2. We observe that all three policies achieve decent success rates, with the hightest success rate up to 68% achieved by the ACT policy. Notice that the task is challenging because the wiping goals are uniformly sampled over the entire table surface. The original UNet-based DP achieves a relatively inferior performance, with a success rate of 56%. However, by adding the transformer backbone, we can achieve a relative improvement of 18%, from a success rate of 56% to a success rate of 66%. We hypothesize that the transformer backbone greatly increases the model expressivity, allowing it to better model complicated whole-body robot behaviors.

Regarding the other metric *contact ratio*, we observe that all three policies maintain contact with the table surface for more than half portion of rollouts. This suggests the effectiveness of our constrained motion generation — by training policies on trajectories with the desired object-contact behaviors, the policies learn to maintain the contact.

Table 2: **Summary of quantitative results.**

Method	**ACT**	**DP**	**DP w/ Transformer**
Success Rate	68%	56%	66%
Contact Ratio	71%	64%	63%

5.2 Qualitative Analysis

We visualize rollouts from three policies as demonstrated in Fig. 4. For the ACT policy (Fig. 4a), although it achieves the highest quantitative performance, we notice that it generates jerky behaviors with discontinued movements. By contrast, we observe that DP's behaviors are smoother. This is potentially due to the fact that simply training whole-body actions with L2 loss cannot produce coherent behaviors. Regarding the UNet-based DP, because it achieves the lowest quantitative

(a) ACT rollout.

(b) DP rollout.

(c) DP with transformer rollout.

Figure 4: **Visualizations of policy rollouts.**

performance, we observe that the robot hand sometimes will not directly wipe to the goal positions, as illustrated in Fig. 4b. Finally, regarding the qualitative behaviors of the DP with transformer variant, it strikes a balance between strong quantitative performance and smooth qualitative behaviors (Fig. 4c).

6 Discussion

From results presented in the previous section, we conclude that the DP with transformer variant achieves the best performance overall, suggesting the effectiveness and helpfulness of our proposed data synthesis method. However, limitations still exist. The first limitation lies in the fact that all experiments are still conducted in the simulation, and transfer the policies trained in simulation to the real world can be challenging. Second, our data synthesis method still relies on manual specification of task-specific constraints. Future work could automate the constraint constrution through LLMs or VLMs.

7 Conclusion

We introduce a method for synthesizing robot demonstrations via constrained motion planning for a table-wiping task. Policies trained on 100,000 such demonstrations perform the task effectively, highlighting the viability of synthetic data for imitation learning. However, our method still has limitations in sim-to-real transfer and the manual specification of task-specific constraints. Future work could focus on automating the constraints construction using LLMs, and tackling the sim-to-real transfer problem.

8 Team Contributions

Yunfan completes the entire project.

Changes from Proposal No change.

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